Evaluating the Disentanglement of Latent Variables by the DISCoVeR Variational Autoencoder Framework

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ABSTRACT

Disentangled representation learning in single cell omics aims to separate condition-specific and condition-invariant factors of variation in single cell RNA sequencing (scRNA-seq) data, facilitating interpretable and biologically meaningful latent representations. DISCoVeR, a recently proposed variational autoencoder framework designed for this purpose, was investigated in this project. The original scRNA-seq DISCoVeR experiment was reproduced to to evaluate DISCoVeR, and the study was extended by training the model on two additional datasets to assess generalizability. The results indicate that while DISCoVeR performs as expected on the original dataset, it exhibits poor generalizability across other scRNA-seq experiments. In contrast, Conditional Subspace VAE, the framework on which DISCoVeR is built, demonstrates robust performance and better generalizability across datasets, suggesting its suitability for broader applications in learning single cell disentangled representations.

1 INTRODUCTION

1.1 Disentangled Representation Learning

Observable data is influenced by certain hidden, unobserved, underlying characteristics referred to as latent variables, which may be too complex to measure directly, but can be estimated by analyzing patterns in the data (Wang et al., 2024). In terms of machine learning, these variables are part of the model, but remain unobserved, and are thus not present in the dataset (Kingma and Welling, 2019). Disentangled representation learning refers to the process in which "machine learning models are designed to obtain representations capable of identifying and disentangling the [latent variables] hidden in the observed data" (Wang et al., 2024). In general, disentangled representations allow models to be more (a) explainable, learning semantically informative and separate latent factors that affect the observed data, (b) generalizable, disentangling representations from the original input and reducing noise, and (c) controllable, allowing the manipulation of disentangled latents to improve data generation (Wang et al., 2024). The goal of disentanglement itself is to keep these latent representations as distinct from each other as possible, by encoding them in independent latent spaces. Current approaches to disentangled representation learning involve variational autoencoders (VAEs), generative adversarial networks and diffusion models (Wang et al., 2024). This project focuses on evaluating the performance of a few VAEs developed for the purpose of learning disentangled representations, while the latter two methods are outside its scope.

1.2 Disentangled Representations in Single Cell Omics

Single cell omics analyzes the molecular profile of individual cells rather than averaging data from whole populations, allowing for an accurate capture of heterogeneity and variable gene expression between cells (Kong et al., 2022; Gunawan et al., 2023). These analyses encompass single cell genomics, transcriptomics, proteomics, epigenomics etc. The development of high-throughput single-cell sequencing technology has led to the emergence and significance of single cell omics in unpacking a "comprehensive genetic map of biological regulatory networks" (Kong et al., 2022). Learning disentangled representations of singe cell omics data facilitates the analysis of the factors underlying cell heterogeneity (Gunawan et al., 2023). This enhances the ability of models to generalize to new data, supporting downstream tasks such as domain adaptation and transfer learning, by assisting the model in knowing what is truly causing changes in its input data (Gunawan et al., 2023). This data could comprise of, for example,

gene expression in various cells obtained from single cell RNA sequencing (scRNA-seq). If we assume the dataset includes a labelled condition variable, disentangled representation learning can aim to separate condition-specific and condition-invariant latent variables. This will facilitate attribute manipulation, which in turn will improve data simulation and generation, by switching between condition labels (Wang et al., 2024; Gunawan et al., 2023).

1.3 Project Aim

The aim of this project was threefold:

(a) To build conceptual knowledge and understanding by extensively reviewing the literature on concepts such as probabilistic graphical models, variational autoencoders (VAEs), and disentangled representation learning in single cell omics, in order to investigate the performance of a new VAE framework named DISCoVeR for the purpose of disentangling condition-specific and condition-invariant latent variables in scRNA-seq data

(b) To reproduce the scRNA-seq experiment conducted by Slavutsky et al. (2025) to evaluate DISCoVeR's performance, and additionally contribute mutual information scores to aid evaluation

(c) To extend the experiments by training the DISCoVeR model on two additional datasets and evaluating its generalizability by comparing its performance to other variational autoencoder frameworks

2 THEORY

2.1 Probabilistic Graphical Models

Probabilistic graphical models (PGMs) provide a principled framework to represent complex joint probability distributions by combining probability theory with graph structure (Koller and Friedman, 2009). In a PGM, random variables are represented as nodes and their conditional dependencies as edges, allowing high-dimensional distributions to be factorized into simpler components (Koller and Friedman, 2009). This structured representation enables explicit inference about causal relationships between variables. Many modern latent variable models, including variational autoencoders, can be represented as directed PGMs, where observed data are generated by (or depend on) latent variables, whose nodes are shaded (Koller and Friedman, 2009). In such graphs, edges are directed from the latent variable to the observed variable it influences. An example PGM is shown in fig 1, which represents the assumed structure of data the models examined by this project are built to learn. Here, the observed data x is caused by two latent variables, z and w, where w is dependent on a condition y, while z is invariant across all labels of y.

2.2 Bayesian Inference

Bayesian inference is a framework for reasoning under uncertainty by combining prior beliefs with observed data (Kingma and Welling, 2019). A prior represents what is believed about unknown variables before seeing data, while the likelihood describes how probable the observed data is given a particular values of those variables (Kingma and Welling, 2019). Together, they define a joint distribution over variables and data, where a joint distribution describes the probability of all random variables in a model being realized together (Kingma and Welling, 2019). Conditioning this joint distribution on the observed data yields the posterior, which updates prior beliefs in light of evidence, and represents the probability of the unknown variables having these particular values, given the observed data (Kingma and Welling, 2019). This posterior is calculated by normalizing the joint distribution over

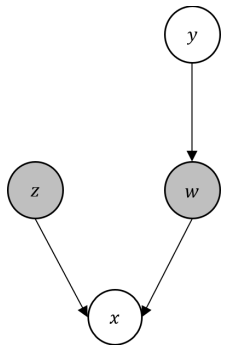

Figure 1. Probabilistic graphical model displaying causal relationships between input x, latent variables z and w, and condition y

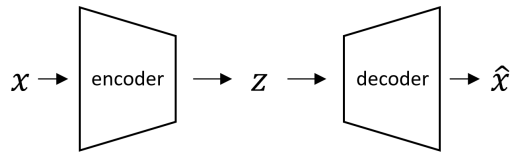

Figure 2. Autoencoder architecture, where x is the input data, z is the latent variable, and $\hat{x}$ is the reconstructed data

the marginal of the observed data, which is the probability of the observed data independent of all other variables (Kingma and Welling, 2019). Equation 1 represents Bayes' theorem, where $p(z \mid x)$ is the posterior, $p(x \mid z)$ is the likelihood, $p(z)$ is the prior, and $p(x)$ is the marginal of the observed data x. The posterior for fig 1 is represented by $p(w, z \mid x, y)$.

$$p(z \mid x) = \frac{p(x \mid z)\, p(z)}{p(x)} \tag{1}$$

2.3 Auto Encoders

Autoencoders are neural network models designed to learn compact representations of high dimensional data by training an encoder–decoder architecture to reconstruct inputs from a lower dimensional latent space (Bank et al., 2021). The encoder maps the input data to a latent representation, while the decoder maps this representation back to the original data as a reconstruction (Bank et al., 2021). A simple pictorial depiction of this process is in fig 2. The latent spaces of these models can be difficult to interpret, as the encoder maps inputs to specific points in the space, and the regions outside these specific points are uninformative. This makes it difficult for decoders to generate new data from different points in the latent space, despite being successful at reconstructing the original input from the specific encoded latent vectors. This motivates the development of variational autoencoders.

2.4 Variational Inference and Variational Autoencoders

Variational inference aims to approximate a posterior distribution which may be too complex to compute on its own (Kingma and Welling, 2019). The goal is for the approximate distribution $q(z \mid x)$ to be as close as possible to the true posterior distribution $p(z \mid x)$. This can be determined by the Kullback–Leibler (KL) divergence between q and p, which is a measure of relative entropy or dissimilarity between two probability distributions (Kingma and Welling, 2019). A lower KL divergence score means q is closely approximating p.

This approach is leveraged by variational autoencoders (VAEs), which map the input data to a probability distribution of the latent variable instead of a singular point, so the decoder can sample values from this distribution and reconstruct the input, as well as generate new output data (Kingma and Welling, 2019; 2022). This solves the problem autoencoders face, where most of their latent space is uninterpretable. A simple pictorial depiction of variational autoencoders is in fig 3. Eqn 2 is the objective that VAEs aim to maximize. The first term represents reconstruction loss, and the second represents the KL divergence between the prior for the latent variable and its encoded approximate distribution (Kingma and Welling, 2019; 2022). Since KL divergence values are always non-negative, the objective is always greater than or equal to the reconstruction term (the evidence). Thus, the objective is called the evidence lower bound (ELBO) (Kingma and Welling, 2019; 2022). Maximizing ELBO maximizes the log likelihood of the reconstructed data from the encoded latent variable and minimizes the KL divergence between the approximate posterior and prior on the latent variable. Fig 4 represents the structure of a VAE based on the PGM in fig 1.

$$\mathcal{L} = \mathbb{E}_{q(z|x)}[\log p(x|z)] - \mathrm{KL}\big(q(z|x) \,\|\, p(z)\big) \tag{2}$$

Figure 3. Variational autoencoder architecture, where x is the input data, z_μ and z_σ are the mean and standard deviation of the latent probability distribution, z is the latent variable sampled from this distribution, and $\hat{x}$ is the reconstructed data

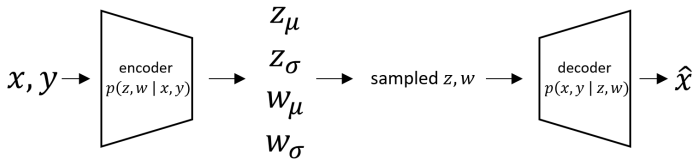

Figure 4. Variational autoencoder architecture where y is a condition variable, w is a condition-specific latent variable, and z is a condition-invariant latent variable

2.5 Baseline VAEs for Disentangled Representation Learning

The objectives of the baseline VAE models used for comparison in Klys et al. (2018) and Slavutsky et al.'s (2025) experiments are as follows:

(a) Conditional VAE:
$\mathbb{E}_{q(z|x,y)}[\log p(x \mid z, y)] - \mathrm{KL}(q(z \mid x, y) \,\|\, p(z)) \leq \log p(x \mid y)$
(b) Conditional VAE with Information Factorization:
$\max_\phi \, \min_\psi \, \mathcal{L}\big(r_\psi(q_\phi(z \mid x)), \, y\big)$
(c) Domain Invariant VAE:
$\mathbb{E}_{q(z,w|x)}[\log p(x \mid z, w)] + \mathbb{E}_{q(w|x)}[\log q(y \mid w)] - \mathrm{KL}(q(z \mid x) \| p(z)) - \mathrm{KL}(q(w \mid x) \| p(w \mid y)) \leq \log p(x, y)$
(d) Characteristic Capturing VAE:
$\mathbb{E}_{q(z,w|x,y)}\left[\frac{q(y|w)}{q(y|x)} \log \frac{p(x|z,w)}{q(y|w)}\right] - \mathrm{KL}(q(z \mid x) \| p(z \mid y)) - \mathrm{KL}(q(w \mid x) \| p(w \mid y)) + \log q(y \mid x)$

2.6 Conditional Subspace VAE

2.6.1 Motivation

Klys et al. (2018) observed that latent spaces of standard VAEs can be hard to interpret or control when considering features correlated to specific labels in the data. To counter this, some models, such as the conditional VAE, learn condition-specific information but do not separately learn condition-invariant information (Klys et al., 2018). In contrast, some models, such as the conditional VAE with information factorization, disentangle the latent space from the condition, but lose condition-specific information (Klys et al., 2018). Thus, the conditional subspace VAE was developed, which explicitly encodes the input data into two separate latent spaces, with the aim of embedding condition-specific information in one space, and condition-invariant information in the other. The hierarchical CSVAE extends the CSVAE architecture for multicondition datasets (Beker et al., 2024).

2.6.2 Architecture and Objective

Fig 5 describes the encoder-decoder architecture for this model. Here, the input comprises of a dataset $D = [(x_1, y_1), ..., (x_N, y_N)]$, where $x \in \mathbb{R}^n$ and $y \in [1, ..., k]$ (Klys et al., 2018). The objective function for CSVAE has two optimizations: (a) minimizing reconstruction loss of x from z, w and (b) minimizing mutual information $I(y; z)$ through an adversarial component (Klys et al., 2018).

The ELBO loss for the reconstruction of x from z, w is as follows, where the expectation term represents the reconstruction loss, and the two KL terms aim to keep the z and w encodings close to their priors (Klys et al., 2018):

$M_1 = \mathbb{E}_{D(x,y)}\big[- \mathbb{E}_{q_\phi(z,w|x,y)} \log p_\theta(x|w, z) + \mathrm{KL}(q_\phi(w|x, y) \| p_\gamma(w|y)) + \mathrm{KL}(q_\phi(z|x, y) \| p(z)) - \log p(y)\big]$

The adversarial term attempts to minimize $I(y; z) = H(y) - H(y \mid z)$ by maximizing conditional entropy $H(y \mid z)$, since the

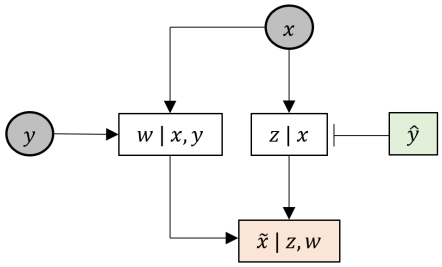

Figure 5. Encoder-decoder architecture for CSVAE, where x is the input data, y is the labelled condition, w and z are the encoded latent variables, $\tilde{x}$ is the reconstructed data from the w and z, and $\hat{y}$ along with the inhibition arrow represents the adversarial component

prior on y is fixed (Klys et al., 2018). This leads to an intractable integral $\iiint_{Z,Y,X} p(z \mid x)\, p(x)\, p(y \mid z)\, \log p(y \mid z)\, dx\, dy\, dz$, that can be made tractable by approximating two distributions, q_δ and q_ϕ, giving the expectation term (Klys et al., 2018):

$M_2 = \mathbb{E}_{q_\phi(z|x)D(x)}\left[\int_Y q_\delta(y \mid z)\, \log q_\delta(y \mid z)\, dy\right]$

M_2 can be computed using a classifier, N, that attempts to maximize predictions of y from the z encoding (Klys et al., 2018):

$N = \mathbb{E}_{q_\phi(z|x)D(x,y)}\left[q_\delta(y \mid z)\right]$

Combining these terms and weighting them gives the final min-max objective for CSVAE in equation 3 (Klys et al., 2018). The model aims to minimize reconstruction loss, as well as minimize mutual information between y and z by penalizing a classifier. The adversarial component is optimized separately in an iterative process, where the classifier tries to make good predictions of y from z, and the encoder tries to encode z so that each time, the best prediction the classifier can make worsens until it is is no better than a random guess (Klys et al., 2018).

$$\min_{\theta,\phi,\gamma}\ \beta_1 M_1 + \beta_2 M_2 \quad \max_{\delta}\ \beta_3 N \tag{3}$$

2.6.3 Prior

The prior is set as follows (Klys et al., 2018):

$p_z = \mathcal{N}(0,1)$

$p(w_i|y=i) = \mathcal{N}(\mu_i, \sigma_i)$, for each y-label i from 0 to k

In their experiments, Klys et al. (2018) assume binary labels of y and arbitrarily assign

$\mu_1 = (0,0), \sigma_1 = (0.1, 0.1), \mu_2 = (3,3), \sigma_2 = (1,1)$, aiming to cluster one label near the origin, and the other away.

2.7 DISCoVeR VAE: DISentangling COmmon and Variant Representations

2.7.1 Motivation

Slavutsky et al. (2025) observed that other disentanglement methods using VAEs depend on "carefully crafted priors, difficult to specify in high-dimensional complex settings like single-cell genomics, or fail to achieve a clean separation" between latent variables. Thus, DISCoVeR was developed, which builds upon the CSVAE architecture by additionally reconstructing x from z alone (Slavutsky et al., 2025).

2.7.2 Architecture and Objective

Fig 6 describes the encoder-decoder architecture for this model. As before, the input comprises of a dataset $D = [(x_1, y_1), ..., (x_N, y_N)]$, where $x \in \mathbb{R}^n$ and $y \in [1, ..., k]$ (Slavutsky et al., 2025). The objective function for DISCoVeR has three optimizations: (a) maximizing reconstruction of x from z, w, (b) maximizing reconstruction of x from z alone, and (c) minimizing mutual information $I(y; z)$ through an adversarial component (Slavutsky et al., 2025).

The ELBO loss for the reconstruction of x from z, w is the same as that for CSVAE, where the expectation term represents the reconstruction loss, and the two KL terms aim to keep the z and w encodings close to their priors (Slavutsky et al., 2025):

$\mathcal{L}_w(q_{w|x,y}, p; x, y) := \mathbb{E}_{q_{z|x}}\Big[\mathbb{E}_{q_{w|x,y}}\big[\log p(x \mid z, w)\big] - D_{\mathrm{KL}}(q_{z|x} \,\|\, p_z) - D_{\mathrm{KL}}(q_{w|x,y} \,\|\, p_{w|y})\Big]$

The ELBO loss for the reconstruction of x from only z is the same as that for a standard VAE (Slavutsky et al., 2025):

$\mathcal{L}_z(q_{z|x}, p; x) := \mathbb{E}_{q_{z|x}}\big[\log p(x \mid z)\big] - D_{\mathrm{KL}}(q_{z|x} \,\|\, p_z)$

The adversarial term is similar to the CSVAE adversarial term, with the key difference being that the classifier, $g(y|z)$, attempts

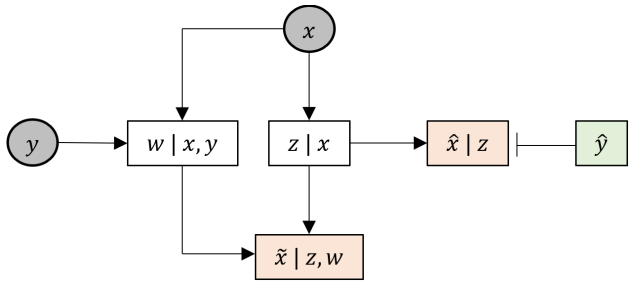

Figure 6. Encoder-decoder architecture for DISCoVeR, where x is the input data, y is the labelled condition, w and z are the encoded latent variables, $\tilde{x}$ is the reconstructed data from the w and z, $\hat{x}$ is the reconstructed data from only z, and $\hat{y}$ along with the inhibition arrow represents the adversarial component

to predict y from the reconstruction of x from z, not from the encoded z itself (Slavutsky et al., 2025). The classifier is trained to minimize the expected negative log likelihood (Slavutsky et al., 2025):

$-\mathbb{E}_{q(z|x)}\big[\log g(y \mid z)\big]$

Combining these terms and weighting them gives the final min-max objective for DISCoVeR in equation 4 (Slavutsky et al., 2025). The model aims to maximize both reconstructions, as well as minimize the classifier's ability to predict y from z. Like in CSVAE, the adversarial component is optimized separately in an iterative process, where the classifier tries to make good predictions of y from the reconstructed x from z, and the encoder tries to encode z so that each time, the best prediction the classifier can make worsens until it is is no better than a random guess (Slavutsky et al., 2025).

$$\max_{q_{z|x}\in\mathcal{Q}_z}\ \max_{q_{w|x,y}\in\mathcal{Q}_w}\ \min_{g_{y|z}\in\mathcal{G}} \mathbb{E}_{p_{x,y}}\big[L_\alpha(q_{z|x}, q_{w|x,y}, g_{y|z}; x, y)\big] \tag{4}$$

2.7.3 Prior

The prior is set as follows (Slavutsky et al., 2025):

$p_z = \mathcal{N}(0, I)$

$p(w|y) = \mathcal{N}(\mu_k, I)$, where $\mu_k := \mathbb{E}_{p_{x|y=k}}\big[\mathbb{E}_{q(z|x)}[z]\big]$

Specifying μ_k as the expected inferred z within each class induces a coupling between z and w through the data distribution, encouraging the z embedding to be informative (Slavutsky et al., 2025). Additionally, if z is truly condition-invariant, μ_k would collapse to a shared mean $\mu = \mathbb{E}_{p_x}\big[\mathbb{E}_{q(z|x)}[z]\big]$ (Slavutsky et al., 2025). Here, $p(w|y)$ is centered at a meaningful point in the latent space rather than an uninformative one, preserving the convex-concave objective structure (Slavutsky et al., 2025).

3 METHODS

3.1 Literature Review

The DISCoVeR framework, along with the code for its experiments, were obtained from v1 of the Slavutsky et al. (2025) paper, provided by the project supervisor, Dr. Jiarui Ding. The papers describing CSVAE (Klys et al., 2018) and HCSVAE (Beker et al., 2024) were obtained from the references of the DISCoVeR paper, while the objectives for DIVA and CCVAE were taken directly from their description in the DISCoVeR paper. Similarly, the objectives for the Conditional VAE and Conditional VAE with Information Factorization were obtained from their description in the CSVAE paper. Conceptual understanding of variational autoencoders was developed using the revised version (v11) of the original variational autoencoder paper (Kingma and Welling, 2022), as well as a monograph on VAEs (Kingma and Welling, 2019). Both sources were obtained from Google Scholar searches. The monograph was also studied to build knowledge on background concepts such as probabilistic models and Bayesian inference. Google Scholar searches also provided sources to develop intuition about autoencoders (Bank et al., 2021), probabilistic graphical models (Koller and Friedman, 2009) and disentangled representation learning in single cell omics (Wang et al., 2024; Gunawan et al., 2023; Kong et al., 2022).

3.2 Datasets

The dataset for experiment 1 was obtained from the GitHub repository for the Slavutsky et al. (2025) paper. The datasets for experiment 2 and 3 were provided as Seurat objects in .rda files by the project supervisor, Dr. Jiarui Ding (original data was from Droin et al. (2021) and Okada et al. (2025) respectively).

All 3 datasets were initialized as AnnData objects, and preprocessing consisted of filtering for cells with a minimum of 200 genes, genes present in a minimum of 3 cells, and 2000 highly variable genes. This filtering was done to reduce noise. The highly variable genes filter was applied following standard approaches for single cell RNA sequencing pipelines.

For experiment 2, the categorical zone feature was computed from the provided continuous zonation feature using MatLab files from the GitHub repository containing the data (Droin et al., 2021). These MatLab files included a matrix of probabilities of each cell belonging to one of the eight spatial zones. The zone label thus assigned to each cell was the zone that it had the highest probability of belonging to according to the matrix.

3.3 Model Implementation and Training

The Slavutsky et al. (2025) implementations were used to train seven VAE models (CSVAE with and without adversarial element, HCSVAE with and without adversarial element, DIVA, CCVAE, and DISCoVeR) on the preprocessed datasets. This implementation used zero-inflated negative binomial models for all seven architectures.

For all experiments, the preprocessed AnnData object was passed into a DataLoader with batch size 128, and a singular random seed of 42 was used. Experiment 1 was a reproduction of the scRNA-seq experiment involving PBMCs from the DISCoVeR paper. In this experiment, models were trained following the code provided in the paper, using the same priors as the paper and the same weights that were derived from hyperparameter tuning performed by Slavutsky et al. (2025). Like the original experiment, training employed early stopping with a patience of 50 epochs, where the patience counter was reset to 0 if the models improved, so the models only terminated after 50 consecutive epochs with no improvement. For experiments 2 and 3, the models were trained in a similar fashion, but with a default value of 1.0 for all weights (z KL weight, w KL weights, reconstruction weight, adversarial weight, and reconstruction from z weight), and a prior consisting of $\mu = [0, 1, ...k]$ and $\sigma = [0.1, 0.1, ...0.1]$ for the w embedding for the 2 CSVAE and 2 HCSVAE models. For these experiments as well, early stopping was used, but with a patience of 30.

For experiment 3, hyperparameter tuning for the CSVAE model was performed in a separate experiment, where the weights were set according to those described in the original CSVAE paper (Klys et al., 2018), and the prior for w was set as $\mu = [0, 1, 2, 3, 4]$, with varying σ: $[0.1, 0.1, 0.1, 0.1, 0.1]$, $[0.2, 0.2, 0.2, 0.2, 0.2]$, $[0.5, 0.5, 0.5, 0.5, 0.5]$, $[0.8, 0.8, 0.8, 0.8, 0.8]$, and $[1.0, 1.0, 1.0, 1.0, 1.0]$.

3.4 Mutual Information Computation

All mutual information results were calculated using software developed by Wally et al. (2025). The distances for categorical labels (cell type and stimulation in experiment 1, zone and time in experiment 2, and time in experiment 3) were computed as 0 if the labels were the same and 1 if they were different. For continuous variables (zone label in experiment 3, and z and w embeddings in all experiments), normalized pairwise euclidean distances were computed.

4 EXPERIMENTS

4.1 Experiment 1

4.1.1 Dataset

The dataset consists of single cell RNA sequencing data from 13,999 peripheral blood mononuclear cells collected from 8 lupus patients. The condition variable y represents stimulation by IFN-β cytokine ($y = 0$ for control and $y = 1$ for stimulated). Thus, the condition-specific latent variable w (categorical) aims to capture this stimulation effect, while the condition-invariant latent variable z (categorical) aims to capture cell-type information.

Fig 7 shows that stimulation is not well separated in the dataset, and although cell type is slightly more well separated, there is still mixing between clusters.

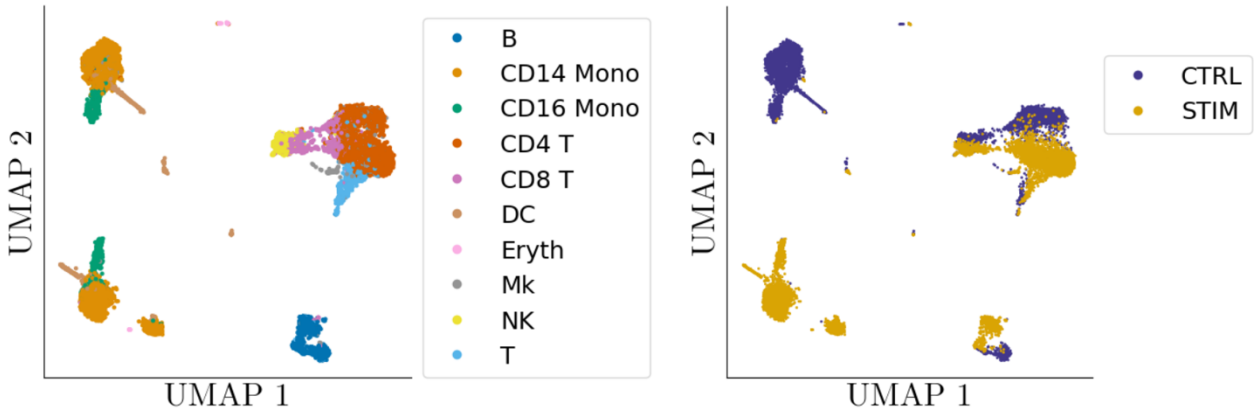

Figure 7. UMAP scatterplots of preprocessed PBMC data from experiment 1 labelled by cell type (left) and stimulation (right)

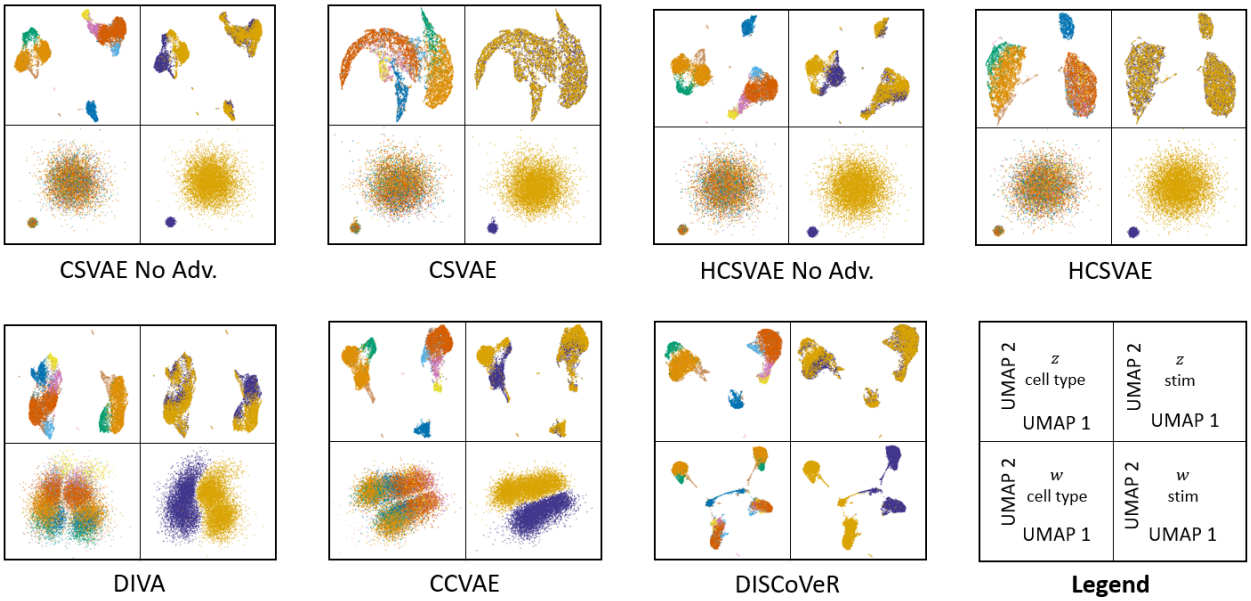

Figure 8. UMAP scatterplots of z (top) and w (bottom) embeddings from 7 VAE models trained on PBMC data from experiment 1, labelled by cell type (left) and stimulation (right)

4.1.2 UMAP Results

The results in fig 8 indicate that:

(a) The z encoding by DISCoVeR captures cell type information very well, while remaining disentangled from the stimulation condition. The w encoding captures distinct clusters for stimulation, but within each cluster, cell type appears to be well separated, indicating low disentanglement between w and cell type.

(b) The z encoding by the CSVAE and HCSVAE models trained without adversary also captures cell type well, but is relatively less disentangled from stimulation than the DISCoVeR z encoding. However, the w encoding by these two models both captures stimulation effect well in isolated clusters, and is very well disentangled from cell type information.

(c) The z encoding by the CSVAE and HCSVAE models trained adversarially capture cell type less well than their non-adversarially trained counterparts, but have extremely high disentanglement from stimulation. The w encodings appear to have similar results as the non-adversarially trained CSVAE and HCSVAE models.

(d) The z encodings by DIVA and CCVAE capture cell type better than the CSVAE and HCSVAE models trained adversarially, but have worse disentanglement than them and DISCoVeR. The w encodings by these two models have distinct clusters for time, but they are less isolated than any of the other models, and cell type information also leaks into these embeddings, signalling low disentanglement.

4.1.3 Mutual Information Results

Fig 9 corroborates the results seen in the UMAP plots:

(a) All models have high mutual information between w and stimulation, with CSVAE and HCSVAE models having the highest. The lowest mutual information between w and cell type (and thus, highest disentanglement) is in the HCSVAE models. The CSVAE models are worse than the HCSVAE models in this regard, but better than the baselines and DISCoVeR, which has the worst disentanglement.

(b) Cell type information is captured well by models, with the non-adversarial CSVAE model performing the best, and disentanglement between z and stimulation is highest in the adversarially trained HCSVAE model, followed closely by the adversarially trained CSVAE model, and then the DISCoVeR model.

(c) DISCoVeR has the lowest disentanglement between w and z, while adversarially trained HCSVAE has the highest, closely followed again by the adversarially trained CSVAE model.

Model	$I(w; stim)$	$I(w; celltype)$	$I(z; stim)$	$I(z; celltype)$	$I(w; z)$
CSVAE no adv.	**0.690767**	0.015867	0.361820	**1.681438**	0.381784
CSVAE	**0.691244**	0.026683	0.009354	1.537786	0.037714
HCSVAE no adv.	**0.690743**	**0.004008**	0.276890	1.630954	0.268547
HCSVAE	**0.690645**	0.008757	**0.007606**	1.378936	**0.015643**
DIVA	0.670366	0.810485	0.358816	1.566723	1.158627
CCVAE	0.673784	0.605179	0.268698	1.630156	1.066973
DISCoVeR	0.681363	1.458018	0.175519	1.574410	2.330181

Figure 9. Mutual information results for 7 VAE models trained on PBMC data from experiment 1

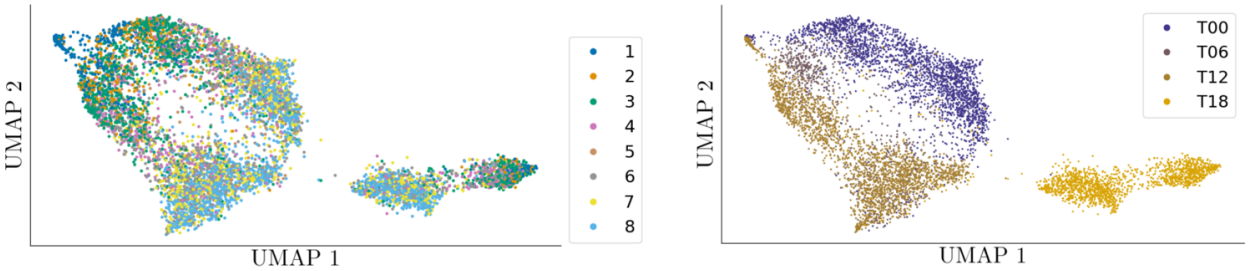

Figure 10. UMAP scatterplots of preprocessed hepatocyte data from experiment 2 labelled by zone (left) and time (right)

4.2 Experiment 2

4.2.1 Dataset

The dataset consists of scRNA-seq data from 7,212 hepatocytes collected from 10 ad libitum fed mice at 4 times throughout the day. The condition variable y represents time of the day ($y = 0, 1, 2, 3$ for zeitgeber time 0h, 6h, 12h, 18h). Thus, the condition-specific latent variable w (categorical) aims to capture this time effect, while the condition-invariant latent variable z (categorical) aims to capture spatial zone information.

Fig 10 shows that time is decently separated into clusters in the dataset, and within each time cluster, zone appears to be decently separated as well.

4.2.2 UMAP Results

The results in fig 11 indicate that:

(a) The z encoding by DISCoVeR captures zone information decently, but captures time information even better, indicating very low disentanglement. The w encoding captures some information about time, but not as well as z or even the original dataset, although it remains disentangled from zone.

(b) The z encoding by the CSVAE and HCSVAE models also capture zone information somewhat decently, with the CSVAE models performing better in this regard. All 4 models encode z that is highly disentangled from time, as well as w that captures distinct isolated clusters for time while remaining highly disentangled from zone.

(c) The z encodings by DIVA and CCVAE do not capture zone information well, but are more disentangled from time information than DISCoVeR's z encoding. These models encode w that capture time information poorer than the CSVAE and HCSVAE models but better than DISCoVeR, but are the most entangled with zone information.

4.2.3 Mutual Information Results

Fig 12 corroborates the results seen in the UMAP plots:

(a) All models have high mutual information between w and time, except DISCoVeR. DISCoVeR has lowest mutual information between w and zone, followed by the non-adversarial HCSVAE and CSVAE.

(b) Mutual information between zone and z is highest in DISCoVeR, followed by the CSVAE and HCSVAE models. Mutual

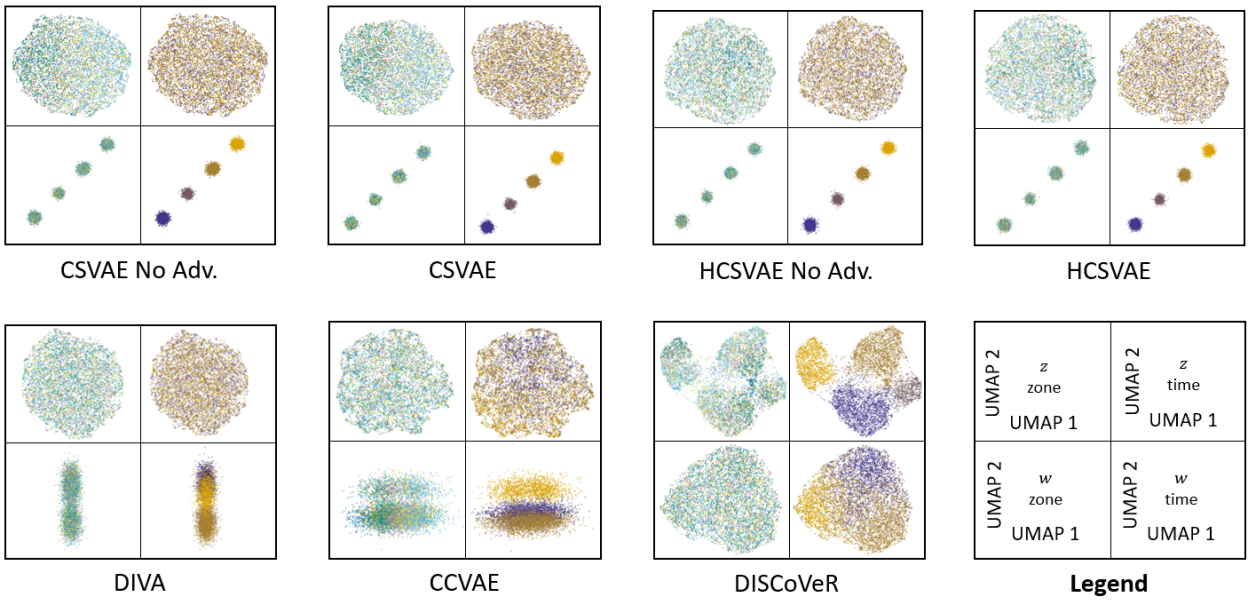

Figure 11. UMAP scatterplots of z (top) and w (bottom) embeddings from 7 VAE models trained on hepatocyte data from experiment 2, labelled by zone (left) and time (right)

Model	$I(w;time)$	$I(w;zone)$	$I(z;time)$	$I(z;zone)$	$I(w;z)$
CSVAE no adv.	**1.318868**	0.009534	0.027627	**0.475197**	0.027958
CSVAE	**1.318868**	0.014864	0.019561	**0.430369**	0.022753
HCSVAE no adv.	**1.318868**	0.001029	0.007264	0.388037	0.007651
HCSVAE	**1.318868**	0.018908	**0.002132**	0.400333	**0.001676**
DIVA	1.318088	0.552224	0.003396	0.001538	0.020169
CCVAE	1.311219	0.549825	0.356368	0.016306	0.329617
DISCoVeR	0.655926	**0.000389**	1.168013	**0.507531**	0.432827

Figure 12. Mutual information results for 7 VAE models trained on hepatocyte data from experiment 2

information between time and z is also highest in DISCoVeR (i.e. lowest disentanglement), and lowest (best disentanglement) in the HCSVAE models.

(c) DISCoVeR has the lowest disentanglement between w and z, followed by CCVAE, while the HCSVAE models have the highest.

4.3 Experiment 3

4.3.1 Dataset

The dataset consists of scRNA-seq data from 18,715 hepatocytes collected from mice at 5 different times after feeding. The condition variable y represents time since feeding ($y = 0, 1, 2, 3, 4$). Thus, the condition-specific latent variable w (categorical) aims to capture this time effect, while the condition-invariant latent variable z (continuous) aims to capture spatial zone information.

Fig 13 shows that time is very well separated into distinct clusters in the dataset (apart from one cluster that seems to be split into two), and within each time cluster, zone appears to be decently separated as well.

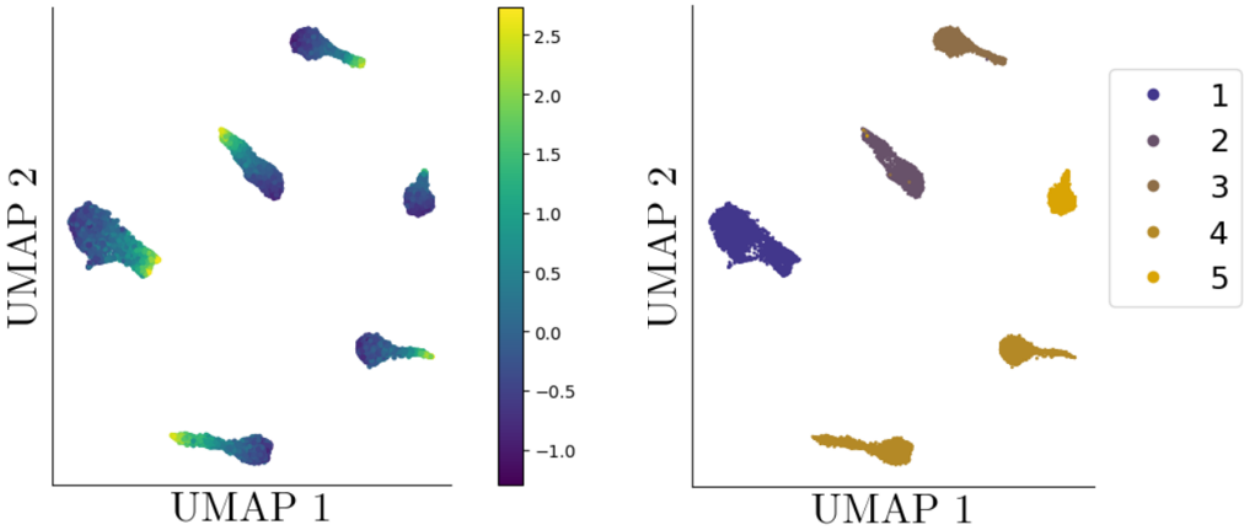

Figure 13. UMAP scatterplots of preprocessed hepatocyte data from experiment 3 labelled by zone (left) and time (right)

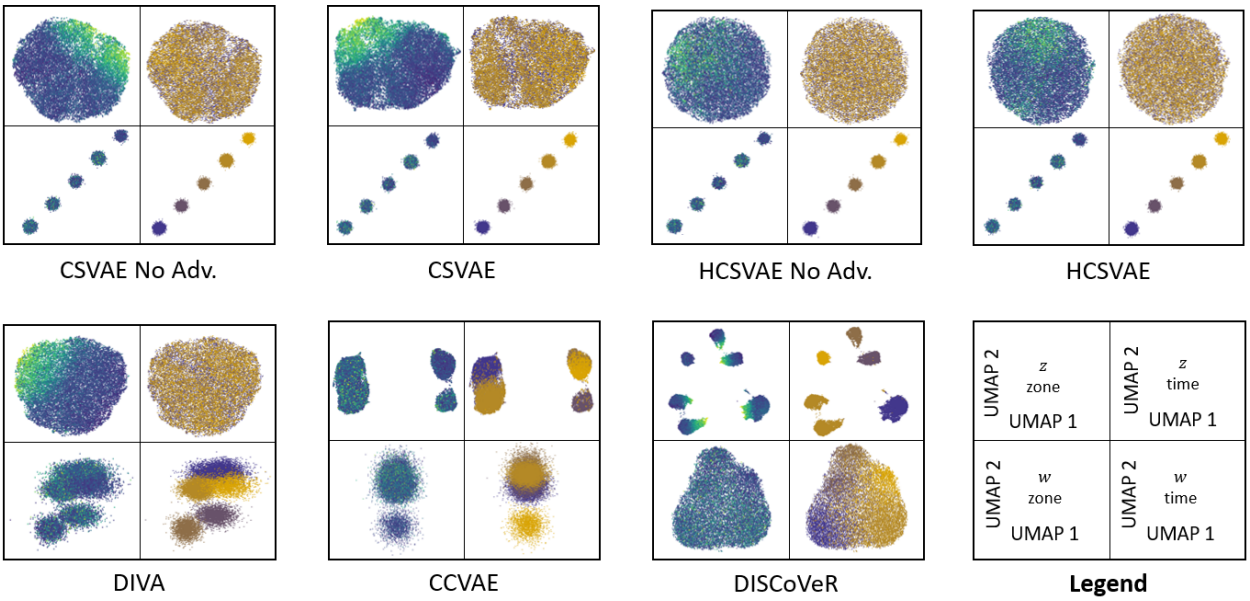

Figure 14. UMAP scatterplots of z (top) and w (bottom) embeddings from 7 VAE models trained on hepatocyte data from experiment 3, labelled by zone (left) and time (right)

4.3.2 UMAP Results

The results in fig 11 indicate that:

(a) The z encoding by DISCoVeR captures time information almost exactly like the original data, with six clusters for the five time labels (one label being split into two clusters), and zone being separated within each cluster. The w encoding does not capture time as well as the original data, as the clusters are not isolated, but remains disentangled from zone information.

(b) The z encodings by the CSVAE models capture zone information well, and are decently disentangled from time information, while those by the HCSVAE models do not capture zone information as well, but are much more disentangled from time. All four models encode w that capture time very well in distinct, isolated clusters, and are well disentangled from zone.

(c) The z encodings by DIVA capture zone well, and is comparably disentangled from time, while the opposite is true for CCVAE. The w encodings by both baselines capture time information better than DISCoVeR, with disentanglement from zone.

Model	$I(w; time)$	$I(w; zone)$	$I(z; time)$	$I(z; zone)$	$I(w; z)$
CSVAE no adv.	**1.494018**	0.067002	0.588156	**0.658192**	0.587002
CSVAE	**1.494018**	0.064868	0.566582	**0.626969**	0.564073
HCSVAE no adv.	**1.494018**	0.061857	0.018303	0.375417	0.020671
HCSVAE	**1.494018**	0.073920	**0.016098**	0.335613	**0.017644**
DIVA	**1.494018**	0.076712	0.071912	0.565387	0.120341
CCVAE	1.488423	0.892668	1.278925	0.087829	1.358537
DISCoVeR	0.849718	**0.022957**	1.474775	**0.771290**	0.714268

Figure 15. Mutual information results for 7 VAE models trained on hepatocyte data from experiment 3

Std. Deviation	$I(w; time)$	$I(w; zone)$	$I(z; time)$	$I(z; zone)$	$I(w; z)$
0.1	1.493864	0.221675	**1.428381**	0.879209	**1.426426**
0.2	1.490770	0.359576	**1.464603**	0.893339	**1.648610**
0.5	1.398316	0.911580	**1.472846**	0.552114	**2.145926**
0.8	1.291382	0.871217	**1.483933**	0.454646	**2.119844**
1.0	1.269006	0.828753	**1.481424**	0.466337	**2.178017**

Figure 16. Mutual information results for hyperparameter tuning of σ for 7 VAE models trained on hepatocyte data from experiment 3

4.3.3 Mutual Information Results

Again, fig 15 corroborates the results seen in the UMAP plots:

(a) All models have high mutual information between w and time, except DISCoVeR. DISCoVeR has lowest mutual information between w and zone, followed by the CSVAE, HCSVAE and DIVA models, all three of which have comparable scores to each other.

(b) Mutual information between zone and z is highest in DISCoVeR, followed closely by the CSVAE models. Mutual information between time and z is also highest in DISCoVeR (i.e. lowest disentanglement), and lowest (best disentanglement) in the HCSVAE models.

(c) CCVAE has the lowest disentanglement between w and z, followed by DISCoVeR, and then the CSVAE models, while the HCSVAE models have the highest.

4.3.4 Hyperparameter Tuning Prior on w

According to fig 16, adversarially training CSVAE models using the weights from the original paper (Klys et al., 2018) resulted in high mutual information (and thus low disentanglement) between the z encodings and time, as well as between the z and w encodings. Changing the standard deviation in the prior for w did not result in much difference in scores between z and time, but produced higher scores between z and w for higher standard deviation values compared to lower.

5 DISCUSSION

DISCoVeR encodes informative latent variables in the first experiment, but achieves lower disentanglement between its w encoding and both cell type and z than was hoped for. The CSVAE and HCSVAE models perform better overall, with good capture of appropriate information by their latent embeddings, and higher disentanglement between the latents encoded by the adversarially trained models. The baselines encode informative latents, but have lower disentanglement than the CSVAE and HCSVAE models.

DISCoVeR has poorer performance in the second experiment, with low disentanglement between its z encoding and both time and w. Although it achieves the highest disentanglement between w and zone, these results are not very encouraging as the

embedding is not very informative anyway, even about time. The CSVAE and HCSVAE models once again perform better overall, with their w and z encodings capturing time and zone well respectively, as well as high disentanglement between the two latents, as well as between each latent and its opposite label. The baselines encode informative w, but it is highly entangled with zone; their z encodings do not capture zone well. For the models that capture zone well within z, the UMAP clusters are not well isolated as spatial zonation within a hepatocyte is truly continuous, and the zone labels were assigned computationally, and may not accurately reflect "true" separation. Cells belonging to two neighbouring zone labels may be highly similar to each other. However, a distinct gradient of colors can be observed, suggesting that the z encoding does group cells in the same zone together, and apart from those in different zones.

In the third experiment as well, DISCoVeR has poor disentanglement results for its z embedding. Once again, despite the high disentanglement between w and zone, its outlook is poor as w is uninformative about time. CCVAE encodes an informative w but has very low disentanglement. The CSVAE, HCSVAE and DIVA models perform better than DISCoVeR and CCVAE, with the HCSVAE models achieving best disentanglement within their z encodings, and the CSVAE models achieving best capture of zone information by their z. Similar to the second experiment, z encodings do not show isolated clusters for zone in the UMAP plots, but a clear gradient of colors can be seen.

DISCoVeR performs decently in the first experiment, but has poor generalizability to the other two datasets. The CSVAE and HCSVAE models outperform it in most ways in every experiment. DIVA performs well in the third experiment, but not in the others, and is thus also not robust. Due to their consistently high performance across experiments, CSVAE and HCSVAE models appear to be the most generalizable and robust overall.

Changing the weights for the adversarially trained CSVAE model to reflect those used in the original paper worsened its disentanglement results in the third experiment. Attempting to hyperparameter tune the standard deviation in the prior for w did not improve results. For the third experiment, it seems that the default value of 1.0 for all weights works better.

5.1 Limitations

Limitations of the experiments are as follows:

(a) A major limitation of the second and third experiments is the use of default values of 1.0 for all weights, as well as arbitrary priors for w in the CSVAE and HCSVAE models. Slight hyperparameter tuning of the standard deviation within the prior for w was attempted for the adversarially trained CSVAE model (using weights from the original paper (Klys et al., 2018)), but this did not improve results. No hyperparameter tuning of weights was performed for any model.

(b) Secondly, the DISCoVeR and CSVAE models appear to be set up to work well with conditions that have binary, categorical labels. This may be a reason they perform better in the first experiment than the second and third experiment, as the second experiment uses a condition with four labels, and the third has a condition with five, continuous labels. This may also affect the prior set on w in DISCoVeR, as it assigns the expectation of z over all condition labels to its mean.

(c) Finally, in the second and third experiment, DISCoVeR's z encoding has extremely low disentanglement from the condition-specific label as well as from w. This may be because the model involves maximizing a reconstruction from z alone, and since the original data captures the condition-specific label well, the z encoding does too.

5.2 Future Work

Future work could focus on hyperparameter tuning the weights for all models (and possibly the prior on w for the CSVAE and HCSVAE models) to test for improved results. It could also involve improving CSVAE and DISCoVeR performance with continous, multi-label conditions. The DISCoVeR framework in specific could possibly be improved further by optimizing its z encoding to remain informative while increasing disentanglement from the condition-specific variable. Models beyond what were used for comparison in the DISCoVeR and CSVAE papers (Slavutsky et al., 2025; Klys et al., 2018) were outside the scope of this project, but it may be worthwhile to evaluate model performance by comparing with other frameworks for disentangled representation learning, both VAE and non-VAE based (such as diffusion models).

6 CONCLUSION

Many disentangled representation learning frameworks seek to separate condition-specific and condition-invariant latent variables in single cell omics data. This improves the generalizability of models, facilitates attribute manipulation, and supports higher quality data generation for analysis. Conditional Subspace VAE and DISCoVeR are two variational autoencoder based architectures developed for this purpose, both of which explicitly encode two distinct latent spaces, and aim to embed all condition-specific information within one space and all condition-invariant information within the other. To achieve this, both frameworks include an

adversarial element that attempts to minimize the mutual information between the condition variable and the condition-invariant latent. DISCoVeR differs from CSVAE by adding a separate reconstruction of the data from its z encoding alone.

Three experiments were performed in this project with single cell RNA sequencing data to evaluate the disentanglement ability of CSVAE and DISCoVer by comparing their performance with each other and baseline mdoels. CSVAE appears to perform the best across all experiments, with its latent variables capturing the appropriate disentangled information well. DISCoVeR seems to have poor generalizability beyond the experiment originally performed by Slavutsky et al. (2025), with high entanglement of condition-specific information in its condition-invariant latent variable, and relatively uninformative encoding of its condition-specific latent variable.

Through this project, the student gained immense knowledge about very relevant probability and deep learning topics that were previously foreign to her, learned to handle RNA sequencing data and work with Pytorch for the first time, developed experimental design skills, troubleshooting abilities, and analytical intution for future computational biology projects, and learned through hands-on experience how theoretical frameworks are translated into functional models, the importance of generalizability and robustness of these models, and methods to evaluate their performance.

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